A Differentiable E-Z Reader: Integrating Neural Language Models into a Cognitive Model of Eye-Movement Control in Reading

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2. We have disclosed all technology that we used in the writing process.
3. We claim responsibility for our submitted work.

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Research Motivation

Eye movements during reading consist of rapid jumps between fixation points (saccades) alternating with brief periods of relative stillness (fixations), and these patterns are directly sensitive to the linguistic properties of the text being processed. Rayner (1998) established that eye movement data reflect moment-to-moment cognitive processing during reading, providing the empirical basis for word-level measures such as first fixation duration (FFD), gaze duration, total reading time (TRT), and skip rate as standard indices of lexical processing difficulty.

The E-Z Reader model (Reichle, Pollatsek, Fisher, & Rayner, 1998) provides a computational account of eye movement control during reading. The model proposes two sequential processing stages for each word: a familiarity check (L1) that initiates saccade programming toward the next word, and a full lexical access stage (L2) that shifts attentional resources. Both stages are computed through assumed parametric functions of word frequency and contextual predictability, inputs that must be obtained from precomputed lexical databases and norming studies. This dependency constrains the model's applicability to corpora for which such norms are available and limits its ability to capture the full range of contextual factors known to influence lexical processing.

A separate line of research has approached reading time prediction from an information-theoretic perspective. Levy (2008) formalized the proposal that processing difficulty during reading is proportional to the surprisal of a word, defined as its negative log-probability given the preceding context. Goodkind and Bicknell (2018) demonstrated empirically that the predictive power of surprisal for reading times scales as a linear function of language model quality, directly linking improvements in language modeling to gains in behavioral prediction. Although these approaches achieve strong predictive performance at the word level, they reduce the model's output to a single scalar per word, which does not permit decomposition of processing difficulty into distinct cognitive stages.

The present study addresses a gap at the intersection of these two traditions. No existing model simultaneously offers word-level reading time prediction from raw text, independence from precomputed lexical norms, and a mechanistically interpretable decomposition of predictions into cognitively grounded processing stages. We propose a differentiable reformulation of E-Z Reader in which the parametric input formulas are replaced by a neural language model trained end-to-end on human eye-tracking data.

Research Question

Can a differentiable reformulation of E-Z Reader, in which lexical processing stages are predicted by a neural language model rather than computed through static parametric equations, achieve word-level reading time prediction accuracy comparable to direct neural regression models, while preserving the mechanistic interpretability of the underlying cognitive architecture?

Background

The E-Z Reader model (Reichle et al., 1998) accounts for eye movement patterns across a range of reading conditions by decomposing word processing into two sequential cognitive events. The familiarity check (L1) is completed rapidly and triggers saccade programming toward the next word. Lexical access (L2) is slower and governs the shift of focal attention. The model generates predictions at the level of aggregated data, fitting mean fixation durations and fixation probabilities across groups of words or experimental conditions. Because its inputs must be supplied from externally normed databases, the model cannot be applied to arbitrary text, and its parameters are typically calibrated by hand to fit a specific corpus.

The theory of surprisal (Levy, 2008) provides a complementary framework for understanding lexical processing difficulty. Under this account, the processing cost of a word is proportional to its negative log-probability given the preceding context, connecting cognitive processing difficulty directly to probabilistic language modeling. Goodkind and Bicknell (2018) formalized this connection empirically, showing that the predictive power of surprisal for reading times is a linear function of language model quality. Wilcox, Pimentel, Meister, Cotterell, and Levy (2023) extended this finding across eleven languages, confirming that the relationship between language model performance and behavioral reading time prediction generalizes beyond English.

Despite this predictive success, surprisal-based approaches do not model the cognitive processes that mediate the relationship between word probability and fixation behavior, yielding a single scalar prediction per word without decomposing processing difficulty into sequential stages. A recent attempt to bridge this divide replaced cloze predictability norms with LLM-derived probabilities as inputs to OB1-reader, a cognitive model of eye-movement control, and found that LLM-based predictability provides a better fit to human eye-tracking data than traditional cloze norms (Lopes Rego, Snell, & Meeter, 2024). However, that approach preserves the non-differentiable simulation architecture and does not allow the cognitive model parameters to be jointly optimized with the language model encoder.

The present study addresses this limitation directly. E-Z Reader is implemented as a discrete event simulation whose operations are not differentiable, precluding gradient-based optimization and joint training with a learned encoder. By reformulating E-Z Reader as a differentiable computation graph and replacing its parametric input functions with the output of a neural language model, both components can be trained simultaneously on human eye-tracking data, enabling the model to learn cognitively structured predictions without relying on external norms.

Methods

Human Subject Experiment

The present study will make use of two existing eye-tracking corpora as the source of behavioral data. The Ghent Eye-Tracking Corpus (GECO; Cop, Dirix, Drieghe, & Duyck, 2017) consists of recordings from 14 monolingual English participants reading a complete novel, yielding approximately 54,000 word-level observations. The Provo Corpus (Luke & Christianson, 2018) contains eye-tracking data from 84 participants reading 55 short expository passages (2,654 words in total), accompanied by cloze-based contextual predictability norms collected from a separate norming sample. Both corpora employed a within-subject design in which all participants read the same materials. The dependent variables are first fixation duration (FFD), gaze duration, total reading time (TRT), and skip rate, measured at the level of individual words. The independent variables are word-level properties that naturally vary across stimuli: orthographic word length, word frequency, and contextual predictability. Behavioral measures will be aggregated across participants per word position and analyzed using word-level correlation metrics, serving as both training targets and evaluation criteria for the machine learning experiment.

Machine Learning Experiment

The GECO corpus will be used for model training and in-distribution evaluation, with sentences partitioned into training, validation, and test sets at the level of complete texts to prevent data leakage. The Provo corpus will be held out entirely as a cross-corpus generalization benchmark, allowing evaluation on text of a different genre and from a different participant population than those seen during training.

Four models will be implemented and compared. The first is the original E-Z Reader (Reichle et al., 1998), using externally supplied word frequency and predictability values as inputs, serving as the cognitive baseline. The second is a differentiable reformulation of E-Z Reader using the same formula-based inputs but with all model parameters optimized via gradient descent, isolating the contribution of end-to-end parameter learning. The third is the full proposed model: a pretrained causal language model serves as an encoder, producing contextual word representations that are projected through three learned output heads predicting L1 processing time, L2 processing time, and skip probability, which are then passed into the differentiable E-Z Reader to produce FFD, gaze duration, TRT, and skip rate predictions. The fourth model is a direct neural regression baseline using the same encoder but with four independent output heads predicting each reading time measure directly, without any cognitive architecture, serving as the critical control for assessing whether the cognitive structure provides any benefit over unconstrained regression.

Model performance will be evaluated using word-level Pearson correlation between predicted and observed values for each dependent variable, computed separately on the GECO held-out test set and on the full Provo corpus. Cross-corpus generalization will be assessed by comparing performance across the two evaluation sets for all four models.

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